

AI Resume Screening & Ranking System

Formal project approach and achieved results

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Executive Summary

This project implements a batch resume screening pipeline for an SDE internship requiring Python fundamentals and practical AI or agentic-system experience. The system processes resumes, extracts structured candidate information, applies hard eligibility rules before ranking, calculates an explainable score, optionally enriches public GitHub data, and produces machine-readable JSON output.

Achieved Results

Metric	Result
Total files	51
Successfully parsed	51
Eligible candidates	43
Rejected candidates	8
Failed resumes	0
Duplicate files	0

System Approach

- Discover PDF, DOCX, and TXT files recursively without modifying the input dataset.
- Extract and normalize text, including clickable PDF hyperlink annotations for GitHub profiles.
- Build conservative structured candidate records with Pydantic models.
- Apply the deterministic Python plus meaningful AI hard filter before scoring.
- Score only eligible candidates using a transparent 100-point model.
- Enrich GitHub activity when enabled, with caching and graceful API failure handling.
- Write ranked candidates, rejected candidates, failures, duplicates, evidence, and summaries to JSON.

Eligibility Logic

A candidate is eligible only when Python evidence and meaningful AI, LLM, RAG, machine-learning, or agentic implementation evidence are both present. Ranking cannot override a failed eligibility condition.

Scoring Model

Category	Maximum	Evidence considered
AI / Agentic / RAG depth	40	Retrieval, embeddings, agents, tools, state, orchestration, evaluation, business logic
Python and backend	30	Python, FastAPI, backend APIs, PostgreSQL, Redis, async, architecture
Cloud / deployment / full stack	15	Cloud, Docker, deployment, CI/CD, React, Next.js
GitHub activity	10	Recent public events and relevant repositories
Engineering depth	5	Tests, caching, queues, observability, concurrency, failures

Top Ranked Candidates

Rank	Candidate	Score	Primary signal
1	V Sree Raghu Vardhan	88	Software Engineer with experience building production AI backend systems using Python, Fas
2	Hari Shanker Sharma	87	Designed, built, and shipped four production AI SaaS platforms spanning AI
3	Shaiksultanasumaiya623	82	deployment (GCP, Docker), and React/TypeScript frontends, with hands-on experience integra
4	Prathamesh	77	Final-year AI/ML Engineering student with hands-on experience building production-grade RA
5	ABHINAV	75	LUNACAL.AI
6	Kiran	74	AI
7	Arjun Kumar	73	Pre-final year CS (AIML & IoT) student with hands-on experience across full-stack developm
8	Thejeshwaar Paasila	73	AI-focused Full-Stack Developer, GenAI Engineer, and Startup Founder with expertise in bui
9	Gayathri Palraj	73	Comfortable with Git-based workflows and shipping real, working software end-to-end rather
10	RAJ THUMAR	72	• Built AI-powered workflow automation by integrating OpenAI, Gemini, RAG, and MCP into ba

Example Resume Output

Each resume produces a structured record with identity, eligibility, matched skills, project summary, GitHub status, score details, and ranking when eligible.

Field	Example value
PDF / source file	candidate_30.pdf
Candidate	V Sree Raghu Vardhan
Eligibility	True
Ranking number	1

Total score	88
Matched skills	Python, FastAPI, Django, PostgreSQL, MongoDB, Redis, Docker, AWS, React, Java, JavaScript, SQL
Strengths	Python implementation evidence; AI/LLM/agentive implementation evidence
Concerns	None recorded

Rejected Candidates

Rejected resumes remain in the final JSON and are not ranked. The following table lists every rejected person and the exact eligibility reason.

PDF	Candidate	Rejection reason
candidate_01.docx	Kartikay Sinha	No evidence of Python implementation
candidate_01.pdf	Kartikay Sinha	No evidence of Python implementation
candidate_02.pdf	SHIVAM RAJ	No evidence of Python implementation
candidate_11.pdf	MD AZIZUL ISLAM	No evidence of Python implementation
candidate_18.pdf	Harshit Kumar Mishra	No evidence of Python implementation
candidate_32.pdf	Mukul Pal	No evidence of Python implementation
candidate_42.pdf	ANJALI PATIL	No evidence of Python implementation; No meaningful AI, LLM, RAG, or agentive evidence
candidate_46.pdf	AABHAR GUPTA	No evidence of Python implementation

Reliability and Explainability

- Each resume is isolated so malformed files do not stop the batch.
- Rejection reasons identify the exact failed rule.
- Score evidence and visible penalties support interview discussion.
- The generated JSON preserves detailed machine-readable output for review.